

CHAPTER- 3

METHODOLOGY

3.1 Hardware and Software Requirements

3.1.1 Hardware Requirements

To ensure smooth functioning of the AI-Driven Crop Disease Prediction and Management System focusing on the banana plant, the following hardware was utilized:

- Processor (CPU): Intel Core i5 (8th Gen) or AMD Ryzen 5 (or higher)
Purpose: To handle data preprocessing, training, and evaluation tasks effectively.
- Graphics Processing Unit (GPU): NVIDIA Tesla K80 / T4 (available onGoogleColab)
Purpose: For speeding up deep learning model training and handling large image datasets.
- RAM (Memory): Minimum 8 GB (Recommended: 16 GB or higher)
Purpose: To support multiple processes like data loading, augmentation, and model training without system crashes.
- Storage:Minimum 50 GB free disk space
Purpose: For storing images, model files, and training logs.
- Internet-Connection: High-speed internet
Purpose: Required for accessing Google Colab, uploading datasets, and saving models and results to cloud storage.

3.1.2 Software Requirements

The following software environment and tools were used for project development:

- Development Platform:
 - Google-Colab
Reason: Provides free access to powerful GPUs for deep learning tasks and allows easy saving and sharing of notebooks.
- Programming Language:
 - Python-3.8+
Reason: Widely used for machine learning, deep learning, and image processing with large support of libraries.
- Python Libraries:
 - TensorFlow – For building and training CNN models like EfficientNet, MobileNetV2, DenseNet, InceptionV3.
 - Keras – High-level API for deep learning model creation and management.
 - OpenCV – For image preprocessing tasks like resizing, normalization, and augmentation.
 - Pandas – For structured data handling and management.
 - NumPy – For mathematical computations and matrix operations.
 - Matplotlib, Seaborn – Used for plotting graphs like accuracy, loss curves during training.
 - Scikit-learn – For evaluating the performance of models using metrics like accuracy, precision, recall, and F1-score.
- Dataset Handling:

- Google-Drive:

Reason: For uploading, organizing, and accessing datasets easily during model training on Colab.